

Statistical Inference Report

Healthcare Patient Portal Evaluation

Inferential Statistics and Hypothesis Testing

1 Objective

A clinic introduced a patient portal with automated appointment reminders. The goal is to determine whether the portal reduces patient no-show rates.

2 Dataset Description

The dataset contains 160 patients:

- Portal Group (Treatment)
- Standard Care Group (Control)

Table 1: Variable Definitions

Variable	Description
patient_id	Patient identifier
group	Portal or Standard Care
baseline_no_show_rate	No-show rate before intervention
post_no_show_rate	No-show rate after intervention
age	Patient age
appointments_6mo	Number of appointments

3 Tasks

1. Paired t-test

Using only the Portal group:

- Compare baseline and post-intervention no-show rates.
- Report the t-statistic and p-value.
- State whether the no-show rate changed significantly.

2. ANCOVA

Fit the model:

$$\text{post_no_show_rate} = \beta_0 + \beta_1(\text{group}) + \beta_2(\text{baseline_no_show_rate}) + \epsilon$$

Report:

- The effect of the treatment group
- The p-value for the group variable
- Adjusted mean no-show rates for both groups

3. Compare Results

In a short paragraph:

- Compare the conclusions from the paired t-test and ANCOVA.
- Explain why a control group is important when evaluating an intervention.

4 Discussion

Answer the following:

1. Did the patient portal reduce no-show rates?
2. Why is ANCOVA more informative than a simple before-and-after comparison?
3. Would you recommend continuing the patient portal? Explain briefly.

5 Deliverables

1. A commented analysis script.
2. Two visualizations:
 - Before vs. after no-show rates
 - Treatment vs. control comparison
3. A short report (2 pages maximum).